Variational and Optimization-based Approaches for Event-Based Vision Applications

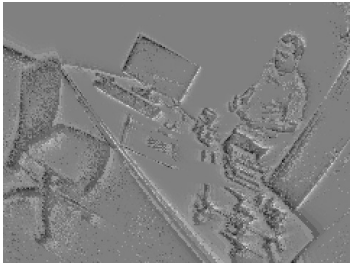
From unobserved modes to well-posed estimators

Daniel Blauvelt

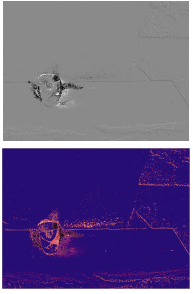
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Dissertation Defense ▪ Fall 2026

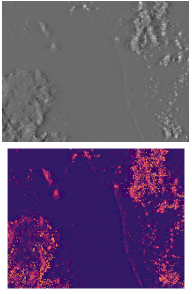
Scenes recovered from brightness changes



Office, handheld; ≈ 2.5M events



Bullet hits a cup; ≈ 350K events

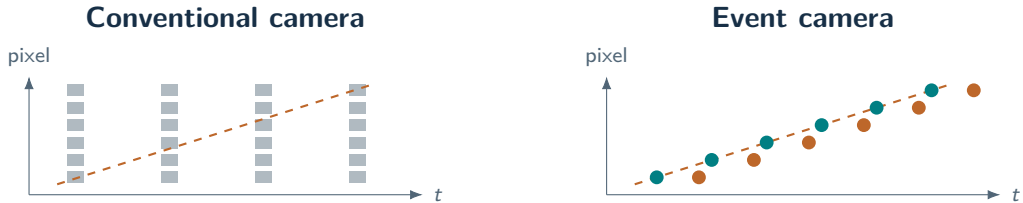


ISS (Falcon Neuro); ≈ 2M events

Cup and ISS: grayscale (top) and false color (bottom) of the same reconstructed frame.

What can brightness changes reveal about images, motion, and depth—and at what computational cost?

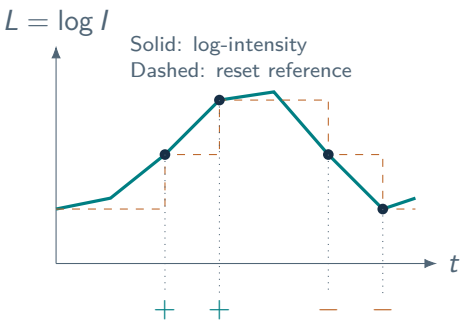
Why an event camera? Each pixel reports its own changes



Dashed line: the leading edge of a moving bright bar. Teal: brightening (+); amber: darkening (−).

- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].

An event camera: each pixel reports threshold crossings



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

The preceding reset s_k^- is **local to that pixel**.

From events to images, motion, and depth

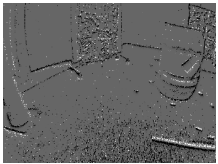
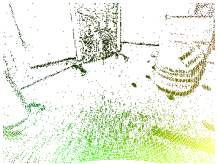


Image reconstruction

Ch. 2

Grayscale: reconstructed value.

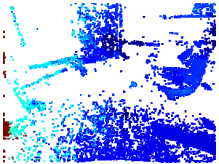


Optical-flow estimate

Ch. 3; event-active pixels



hue: direction
brightness: size
(full at ≈ 14 px)



Stereo-depth estimate

Ch. 4

depth (jet), near to far

0.5 — 8 m

MVSEC [35] indoor_flying2, drone-mounted event cameras, near 47.6 s; stereo depth uses both.

Three estimators, each well posed

Reconstruction (Ch. 2)	One tridiagonal solve per pixel; unique and stable for every positive weight.
Optical flow (Ch. 3)	Three event representations, one quadratic theory; unique flow when edges span two directions.
Stereo depth (Ch. 4)	A unique inner solve at each candidate depth; competitive MVSEC accuracy with the most scored points.

The common question: which quantities do the events determine, and which does the model supply?

Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times t_i are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

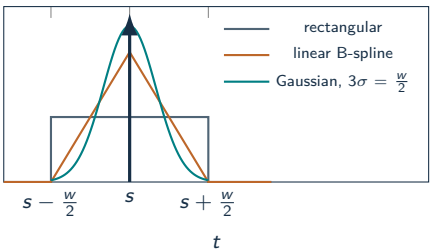
$$\mathbf{A}\mathbf{f} \approx \mathbf{y}.$$

$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

Events determine increments, leaving a constant free.

Events fix the increments; the penalty chooses the reference level.

Temporal kernels: how far each event's influence spreads



Arrow: delta kernel (pure binning). Common support width w; schematic.

Each kernel has unit mass, is a translate, and has compact support:

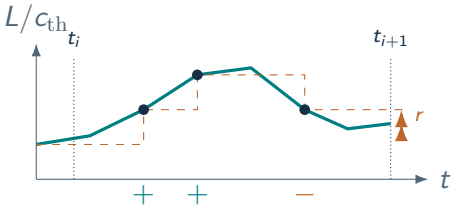
$$\int_{\mathbb{R}} \phi_s = 1, \qquad \phi_s(t) = \phi_0(t - s).$$

Each event splits its sign p_k across the bins it overlaps:

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt, \quad \sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

From threshold crossings to an approximate observation



Solid: L/c_{th}. Dashed: reference at the last event.

Ideal sensor: each event marks a log-brightness change of about one threshold c_{th}.

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt \;\approx\; \frac{L(t_{i+1}) - L(t_i)}{c_{\mathrm{th}}}.$$

Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}}.$$

A unique, stable estimate for every positive weight

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \tfrac{1}{2}\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \tfrac{1}{2}\|\mathbf{\Lambda}\mathbf{f}\|^2, \qquad \mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succeq \lambda_{\mathrm{min}}^2\mathbf{I}.$$

- **Unique:** the normal equations $\mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}$ have exactly one solution.
- **Stable:** $\|\delta\mathbf{f}^*\| \le (2/\lambda_{\mathrm{min}}^2)\,\|\delta\mathbf{y}\|$, so one spurious event moves a pixel's history by at most $2/\lambda_{\mathrm{min}}^2$.
- **Reference level:** $\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$; quiet intervals carry the largest weights and set the level.

One tridiagonal solve per pixel

$\mathbf{A}^\top\mathbf{A}$ has diagonal $(1, 2, \ldots, 2, 1)$ and off-diagonals -1, so

$$\mathbf{K}_{\text{time}} = \mathbf{A}^\top\mathbf{A} + \boldsymbol{\Lambda}^\top\boldsymbol{\Lambda}$$

is tridiagonal and diagonally dominant.

- Thomas algorithm: $O(N)$ operations per pixel, no pivoting.
- Pixels are independent; a pixel without events returns $\mathbf{f} = 0$.
- The newest frame comes from the forward sweep alone: constant work per pixel for each new interval.

Cost on one CPU core
processing time ÷ recording time (below 1 keeps up); float32

Data	Events/s	Batch	Stream	State
DAVIS346	0.25 M	0.04	0.39	1.4 MB
Cup	2.3 M	1.0	1.3	4.9 MB

Batch: 100 ms windows, $N = 50$. Stream: a new frame every 2 ms. State: four numbers per pixel.

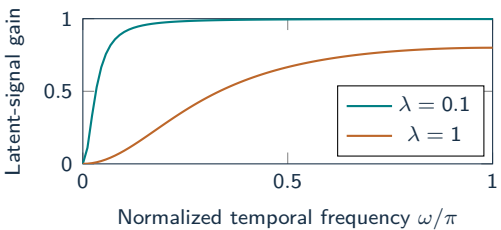
Skipping pixels without events: 10× less work on DAVIS (0.43 → 0.04).

MVSEC indoor flying 1: dense solve 1.5 s → tridiagonal 0.03 s; agrees with MATLAB within 10^{-15}.

What the penalty does to temporal information

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



The baseline is removed.

Low temporal modes are attenuated most.

The penalty selects the baseline; the events leave it free.

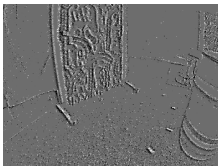
In the images: slow brightness drift fades; passing edges remain.

MVSEC comparison with Antil–Sayre bilevel

MVSEC [35] indoor flying 2



Reference



Event only

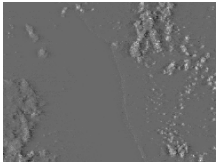
This chapter: events only → activity weights $\mathbf{\Lambda}(dc)$ → fixed convex solve.

Antil–Sayre [2]: events and frames → outer optimization over the blur-kernel parameter z.

	PSNR (dB)		CN	
Sequence	Bilevel	Events	Bilevel	Events
Indoor	15.37	10.85	39755	25.51
Outdoor	8.41	8.71	39755	27.10

PSNR: similarity to a reference frame (dB). CN: condition number of each method's inner linear system.

Shared motion couples temporal profiles to one scene



Each pixel on its own



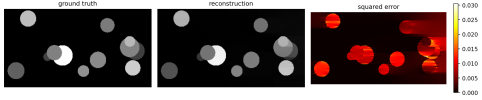
All views, one scene (ESAR)

Chapter 2: each pixel has its own temporal profile.

Shared scene: known motion P maps one static scene θ into every view.

$$\underbrace{B_{\mathrm{SAR}}}_{\text{scene-to-event operator}} = (A \otimes I)\, P.$$

Falcon Neuro on the ISS, near-nadir pass [38]; reconstructions from the ESAR paper [45].



Simulated (v2e [46]), window shifts 1 px per view: scene, shared-scene estimate, squared error.

The aperture problem: what a local view reveals

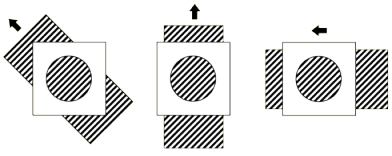
An **aperture** is a restricted observation window: a small view of an edge [66].

Through the windows



The visible stripe motions look the same.

Reveal the full patterns

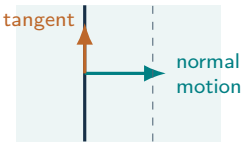


The patterns translate in different directions.

Publisher animations, unchanged: Wolfe et al. [109], Activity 8.2, Movies 5–6.

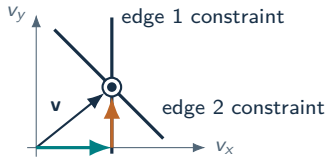
Motion introduces a new invisible direction

Image: one moving edge



A straight edge reveals its normal speed.

Velocity: what the events allow



Edge 1 fixes the normal part; edge 2, if it shares the motion, fixes the tangent part.

Flow model: events give each edge's normal speed; neighbors moving alike supply the rest.
Requirement (main theorem): edges in two orientations somewhere, so their constraint lines cross.

Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

In scene terms: $\mathbf{B}$ records which motion directions the edges measure, and how strongly; b carries the measured values; the α term asks neighbors to move alike.

The main flow theorem: global information controls constants

Unified well-posedness theorem

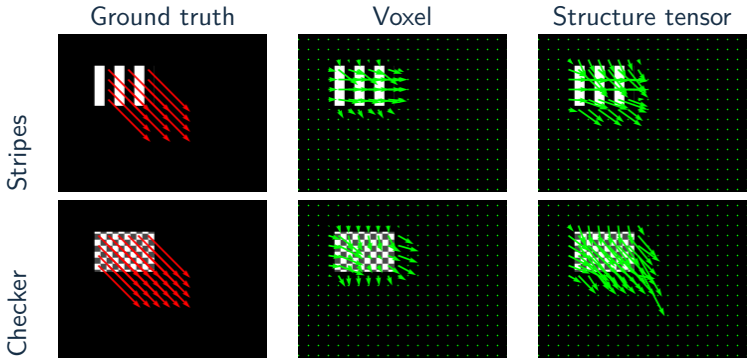
Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

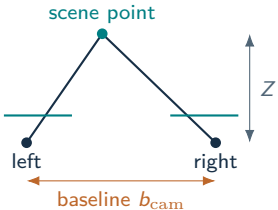
Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

Nonparallel edges control constant velocities; smoothness supplies motion where $\mathbf{B} = 0$.

Known-motion examples make the aperture problem visible



Stereo depth: competitive accuracy, most scored points



$$Z = \frac{f_{\mathrm{cam}} b_{\mathrm{cam}}}{d}, \qquad d = x_L - x_R > 0.$$

Each candidate d pairs the two views, and one shared temporal profile is fitted to both. For fixed d and frozen weights the inner solve is unique.

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	**2.66**
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20

MVSEC indoor flying 1–4; each method scored on its own estimates in $[0.5, 8]$ m. Three more methods in the backup table.

Research opportunities

Sensor modeling

Measure the error terms of slide 9 on real sensors; the stability bounds turn them into estimate bounds.

Learned weights

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{f}^*(\mathbf{\Lambda})$ is smooth.

Shared scenes

Find the motion-invariant modes of B_{SAR} for a trajectory (ESAR [45]); apply to orbital streams.

Edge hardware

Update only pixels with events; measure time and power on a Raspberry Pi 5 and edge sensors.

What the data constrain and what the models prove

Problem	What the observations constrain	What the mathematical model establishes
Temporal reconstruction	Represented brightness increments	Unique regularized history for each fixed positive weight matrix, with perturbation bounds
Optical flow	Event-derived directional motion constraints	Unique H^1 minimizer under integrated identifiability
Stereo depth	Paired temporal event-feature observations	Unique frozen-weight inner solve; the disparity comes from a separate comparison of candidates

Takeaway. Under the stated hypotheses, regularization makes each estimator determinate; accuracy also depends on the representation.

Which directions do the observations control, and which are selected by the model?

Thank you.

Discussion guide

Further detail: time-surface regression and contrast maximization; the full stereo arc (Chapter 4); references.

Observation and temporal inversion

- Minimal uniqueness condition / reference level
- Measure-valued continuous-time limit
- Signal gain versus noise gain
- Metrics; real time; problem size
- Kernels; $\mathbf{\Lambda(y)}$ stability

Flow analysis

- Weak Euler equation and boundary condition
- Full proof, stability, and discrete sweeps
- Exact block-Jacobi spectral argument
- Anisotropy; $\lambda, \alpha \to 0$; event-free regions
- Regression, projection, evaluation; splines
- Straight-edge aperture model

Stereo analysis

- Chapter 4 results summary
- Mean-zero conditioning
- Cost ellipsoid and observation perturbations
- IRLS contraction proof
- Fixed points versus likelihood stationarity
- Implemented scoring and aggregation

Evidence and interpretation

- Depth coverage; alignment; sensitivity
- One lemma behind the uniqueness results
- Contribution; parameters; learning
- Literature anchors and source map

Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

The event measure has a jump primitive

At one pixel, set $\mu = \sum_j p_j \delta_{s_j}$, and choose a consistent right-continuous primitive

$$g(t) = f_0 + \mu((t_0, t]), \qquad g(b) - g(a) = \mu((a, b]).$$

For a standard smooth compactly supported unit-mass kernel,

$$e_h = \mu * \phi_h, \qquad g_h(t) = f_0 + \int_{t_0}^{t} e_h(\xi)\, d\xi.$$

For finitely many interior events, $g_h \to g$ in L^1 as $h \downarrow 0$.

- g belongs to BV; a nonzero interior jump prevents $g \in H^1$.
- The displayed g is an unregularized cumulative-event surrogate.

Takeaway. Keep interval conventions consistent and use increments in $\mathbf{Af} \approx \mathbf{y}$.

Dissertation: Continuous-time formulation (p. 21); endpoint convention made explicit here.

Latent-signal gain and observation-noise gain are different

For the unweighted monocular model with scalar $\lambda > 0$,

$$\mathbf{f}^* = (\mathbf{A}^\top \mathbf{A} + \lambda^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y}.$$

Input	Output modal gain	Interpretation
Latent signal	$g_k/(g_k + \lambda^2)$	Low modes attenuated most
Difference observation	$\sqrt{g_k}/(g_k + \lambda^2)$	Noise amplitude transfer
White difference noise	$\sigma^2 g_k/(g_k + \lambda^2)^2$	Reconstructed modal variance

$$\max_{g \geq 0} \frac{g}{(g + \lambda^2)^2} = \frac{1}{4\lambda^2}.$$

Dissertation: Theorem 4.6.21, Proposition 4.6.23.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Conditioning and reconstruction bias are separate properties.
- Reference-image scores also measure the baseline the model selects.

Dissertation: Table 2.4, Table 2.5.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of a specified matrix	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

Can the reconstruction run in real time?

CPU measurements (one HPC CPU node; float32; processing time ÷ recording time)

Data	Pixels	Events/s	Batch: 100 ms, $N = 50$ 1 thr.	8 thr.	Streaming: 2 ms steps 1 thr.	8 thr.
DAVIS346 pendulum	90k	0.25 M	0.04	0.008	0.39	0.06
Cup	307k	2.3 M	1.0	0.19	1.3	0.21

MVSEC: tridiagonal 0.03 s vs dense 1.5 s; agrees with MATLAB, and streaming with batch, within 10^{-15}.

Why it is fast

- $O(N)$ tridiagonal solve per pixel.
- Skipping pixels without events: 10× less work on DAVIS.

Small and portable

- State: 16 bytes per pixel (1.4 MB DAVIS, 4.9 MB cup).
- NumPy and Numba: the code targets ARM boards.

Takeaway. One core streams DAVIS data at 0.39 of real time with 1.4 MB of state.

Source: edge implementation, October 2026 CPU timings and MATLAB comparison; problem size and deployment options: Backup 83.

The weak equation supplies the natural boundary condition

For every $\boldsymbol{\eta} \in H^1(\Omega;\mathbb{R}^2)$,

$$\int_\Omega (\mathbf{B}\mathbf{v}^* - b)\cdot\boldsymbol{\eta} + \alpha^2 \int_\Omega \nabla\mathbf{v}^* : \nabla\boldsymbol{\eta} = 0.$$

Interior tests give the distributional equation

$$-\alpha^2\Delta\mathbf{v}^* + \mathbf{B}\mathbf{v}^* = b.$$

Since the coefficients imply $\Delta\mathbf{v}^* \in L^2$, Green's identity gives

$$\partial_n\mathbf{v}^* = 0 \quad \text{in } H^{-1/2}(\partial\Omega;\mathbb{R}^2).$$

Takeaway. The theorem uses H^1 fields and weak traces on a Lipschitz domain.

Dissertation: Theorem 3.3.9, Theorem 3.4.2, and Theorem 3.4.3.

Proof: control the mean and the fluctuation separately

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

$$\|\mathbf{w}\|_{L^2} \le C_P \, \|\nabla \mathbf{v}\|_{L^2} .$$

Let $\beta_B = \|\mathbf{B}\|_{L^\infty}$ and $m = \lambda_{\min}(\int_\Omega \mathbf{B}) > 0$ (how well the edges cover every direction).

$$m|\bar{\mathbf{v}}|^2 \le 2 \int_\Omega \mathbf{v}^\top \mathbf{B} \mathbf{v} + 2\beta_B C_P^2 \, \|\nabla \mathbf{v}\|_{L^2}^2 .$$

Therefore the bilinear form satisfies

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) = \int \mathbf{v}^\top \mathbf{B} \mathbf{v} + \alpha^2 \int |\nabla \mathbf{v}|^2 \ge c_0 \, \|\mathbf{v}\|_{H^1}^2 , \qquad c_0 = c_0(\Omega, \alpha, \beta_B, m).$$

Takeaway. Poincaré controls fluctuations, data control the mean, and Lax–Milgram supplies the unique weak solution.

Dissertation: proof of Theorem 3.6.1; Lax–Milgram [72].

Stability follows by subtracting weak equations

Suppose two coefficient pairs share bounds

$$\|\mathbf{B}_i\|_{L^\infty} \le \beta_B, \qquad \lambda_{\min}\left(\int \mathbf{B}_i\right) \ge m > 0, \qquad \|b_i\|_{L^2} \le R_b.$$

Here $c_0 = c_0(\Omega, \alpha, \beta_B, m)$ is independent of b_i. For $\mathbf{z} = \mathbf{v}_1^* - \mathbf{v}_2^*$,

$$\mathcal{B}_1(\mathbf{z}, \mathbf{z}) = \int (b_1 - b_2) \cdot \mathbf{z} + \int \mathbf{z}^\top (\mathbf{B}_2 - \mathbf{B}_1)\mathbf{v}_2^*, \quad \text{hence}$$

$$\|\mathbf{z}\|_{H^1} \le c_0^{-1} \|b_1 - b_2\|_{L^2} + c_0^{-2} R_b \|\mathbf{B}_1 - \mathbf{B}_2\|_{L^\infty}.$$

In scene terms: jitter, binning, and noise events change the coefficients; constants grow as m shrinks (stripes, slide 19). **Accuracy:** the bound concerns the minimizer of the chosen energy; its distance from the true motion depends on coefficient accuracy and on H^1 smoothing across motion boundaries (known-motion AEE, page 76).

Takeaway. The estimate is Lipschitz in its coefficients under uniform identifiability.

Dissertation: Theorem 3.6.1, Eq. 3.61.

Discrete flow: connectivity and sampled directions matter

Fix $\alpha > 0$, $q_i > 0$, and $\mathbf{B}_i = \mathbf{B}_i^\top \succeq 0$ on a connected graph with $w_{ij} = w_{ji} \geq 0$ and degree $d_i = \sum_j w_{ij} > 0$.

One simultaneous block-Jacobi sweep

$$\underbrace{(q_i \mathbf{B}_i + \alpha^2 d_i \mathbf{I}_2)}_{\text{local } 2\times 2 \text{ matrix}} \mathbf{v}_i^{(k+1)} = q_i b_i + \alpha^2 \sum_j w_{ij} \mathbf{v}_j^{(k)}.$$

- At each node, solve for two velocity components using local data and the previous neighbor velocities.
- Update every node from the same old field; then repeat the sweep.
- Scene terms: each pixel blends its edge measurement with neighbors' motion.

If $\sum_i q_i \mathbf{B}_i \succ 0$, the discrete energy has a unique minimizer, and these sweeps converge to it from every initial field.

Dissertation: Theorem 3.7.1 and Corollary 3.7.2; extension to positive node factors q_i.

Why block-Jacobi converges for this graph model

Absorb q_i into $\widehat{\mathbf{B}}_i = q_i \mathbf{B}_i \succeq 0$. Set

$$D_i = \alpha^2 d_i \mathbf{I} + \widehat{\mathbf{B}}_i, \qquad R_{ij} = \alpha^2 w_{ij} \mathbf{I}.$$

For symmetric nonnegative edge weights,

$$\mathbf{z}^\top (D \mp R)\mathbf{z} = \sum_i \mathbf{z}_i^\top \widehat{\mathbf{B}}_i \mathbf{z}_i + \alpha^2 \sum_{\{i,j\}} w_{ij} \left\| \mathbf{z}_i \mp \mathbf{z}_j \right\|^2 .$$

Connectedness and $\sum_i \widehat{\mathbf{B}}_i \succ 0$ eliminate both constant and alternating constant zero modes (the latter on bipartite graphs).

$$D - R \succ 0, \quad D + R \succ 0 \quad \Rightarrow \quad -\mathbf{I} \prec D^{-1/2} R D^{-1/2} \prec \mathbf{I}.$$

Takeaway. The Jacobi iteration matrix is similar to a symmetric matrix with spectral radius strictly below one.

Dissertation: extension of Theorem 3.7.1 and Corollary 3.7.2 by absorbing q_i into $\widehat{\mathbf{B}}_i$. $\alpha > 0$, $d_i > 0$.

Anisotropy needs a uniform ellipticity floor

For the normalized Nagel–Enkelmann tensor [67], write $g = \nabla \rho_{\mathrm{evt}}$ and $\gamma_{\mathrm{NE}} > 0$:

$$\mathbf{D} = \frac{g^{\perp}(g^{\perp})^{\top} + \gamma_{\mathrm{NE}}^2 \mathbf{I}}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

Its tangent and normal eigenvalues are

$$1, \qquad \frac{\gamma_{\mathrm{NE}}^2}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

If $g \in L^{\infty}$,

$$\theta \mathbf{I} \preceq \mathbf{D} \preceq \mathbf{I}, \qquad \theta = \frac{\gamma_{\mathrm{NE}}^2}{\|g\|_{L^{\infty}}^2 + \gamma_{\mathrm{NE}}^2} > 0.$$

Takeaway. Diffusion across event-density edges is reduced, while coercivity survives with $\alpha^2\theta$.

Dissertation: normalized variant of Eq. 3.46, Lemma 3.4.6, Corollary 3.6.4.

A variable diffusion tensor must stay inside the divergence

For differentiable $\mathbf{D}$,

$$\nabla\cdot(\mathbf{D}\nabla u) = D_{11}u_{xx} + 2D_{12}u_{xy} + D_{22}u_{yy} \\ + (\partial_x D_{11} + \partial_y D_{12})u_x + (\partial_x D_{12} + \partial_y D_{22})u_y.$$

Take $\rho_{\mathrm{evt}} = x^2/2$, $\gamma_{\mathrm{NE}} = 1$, and $u = x$:

$$\mathbf{D} = \operatorname{diag}\left(\frac{1}{1+x^2}, 1\right), \qquad \nabla\cdot(\mathbf{D}\nabla u) = \frac{-2x}{(1+x^2)^2} \not\equiv 0.$$

All second derivatives of u vanish, so the principal part alone gives the wrong answer.

Dissertation: Anisotropic numerical discretization (p. 83).

The regression radius formula is a heuristic

For full disks on a square lattice of spacing h, with r/h large,

$$Q \sim \frac{\pi r^4}{4h^2}\mathbf{I}, \qquad U_h(r) = \frac{4}{81}L_H^2 r^4 + \frac{8}{\pi}\sigma^2 h^2 r^{-4}.$$

The minimum of this leading expression, when all parameters are positive, is

$$r^* = (162/\pi)^{1/8}(\sigma h/L_H)^{1/4}.$$

- The full disk must lie in a region with the asserted smoothness bound.
- U_h is an asymptotic expression for the finite-design bound.
- Affine time surfaces have zero curvature bias; this model supplies no finite balancing radius there.

Dissertation: Proposition 3.3.7 and Radius selection as a heuristic.

Uniform and OLS weights agree through quadratics and differ on cubics

The continuous OLS temporal slope over $[s, s+T]$ is

$$\widehat{\beta} = \int_s^{s+T} w_s(t) V_t(t)\, dt, \qquad w_s(t) = \frac{6(t-s)(s+T-t)}{T^3}.$$

Its transport-consistent spatial partner is the same w_s-weighted projection. Pairing it with a uniform projection gives residual

$$\mathbf{v} \cdot \nabla \int_s^{s+T} (T^{-1} - w_s(t)) V(\mathbf{x}, t)\, dt.$$

For $V(x,t) = (2 + x - t)^3$, $\mathbf{v} = (1, 0)$, $s = 0$, $T = 1$, this residual is $1/10$.

Takeaway. Derivative consistency needs its own check, beyond well-posedness of a fixed-coefficient energy.

Dissertation: Section 3.4.1, consistent derivative-operator discussion.

Mean-zero conditioning removes the unobserved mode

In the unweighted scalar model, restrict to

$$\mathcal{V}_0 = \{(\mathbf{f}_L, \mathbf{f}_R) : \mathbf{1}^\top \mathbf{f}_L = \mathbf{1}^\top \mathbf{f}_R = 0\}.$$

The sum–difference transform preserves this space. Its remaining eigenvalues are

$$g_k + \lambda^2, \qquad g_k + \lambda^2 + 2\gamma, \qquad k = 1, \dots, n-1.$$

Thus

$$\kappa_2(\mathbf{M}|_{\mathcal{V}_0}) = \frac{g_{\max} + \lambda^2 + 2\gamma}{g_1 + \lambda^2} \le \frac{g_{\max} + 2\gamma}{g_1}, \qquad g_1 = 4\sin^2 \frac{\pi}{2n}.$$

Takeaway. The restriction is SPD even at $\lambda = 0$. The uniform-in-λ bound depends on the temporal grid; full-space perturbations remain different.

Dissertation: Corollary 4.6.17; Proposition 4.6.18 (notes).

The selection theorem has an exact ellipsoid interpretation

Let

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top), \quad \mathbf{T} = \tfrac{1}{4}\mathbf{A}[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}]\mathbf{A}^\top, \quad \mathbf{P} = \mathbf{S} + \mathbf{T} \succ 0.$$

Assume the ideal match $\mathbf{y}_{d_*} = \mathbf{y}_L$. For $\mathbf{e}_d = \mathbf{y}_L - \mathbf{y}_d$,

$$C(d) - C(d_*) = \|\mathbf{e}_d\|_{\mathbf{P}}^2 - 2\mathbf{y}_L^\top \mathbf{S}\mathbf{e}_d = \|\mathbf{y}_d - \mathbf{c}\|_{\mathbf{P}}^2 - \rho^2,$$

$$\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L, \qquad \rho^2 = \mathbf{y}_L^\top \mathbf{S}\mathbf{P}^{-1}\mathbf{S}\mathbf{y}_L.$$

Takeaway. The ideal match wins if every competitor is outside the ellipsoid. In particular, $\|\mathbf{e}_d\|_{\mathbf{P}} > 2\rho$ is sufficient.

Dissertation: Proposition 4.6.8(a), (b2). Candidate-local unweighted model.

An observation perturbation bound can protect a disparity winner

Keep $\mathbf{y}_L$, the candidate set, $\mathbf{A}$, $\mathbf{\Lambda}$, and γ fixed. Perturb each right observation by $\boldsymbol{\xi}_d$. The quadratic candidate cost has

$$\nabla_{\mathbf{y}_d} C = 2\mathbf{P}(\mathbf{y}_d - \mathbf{c}), \qquad \nabla^2_{\mathbf{y}_d} C = 2\mathbf{P}, \qquad \|\mathbf{P}\| \leq \frac{1}{2}.$$

Consequently a uniform bound is

$$\eta = \max_d \left(\|\mathbf{y}_d - \mathbf{c}\| \, \|\boldsymbol{\xi}_d\| + \frac{1}{2} \|\boldsymbol{\xi}_d\|^2 \right).$$

If $\eta < \Delta/2$, the unique winning disparity is preserved.

Takeaway. The ideal correct match may be perturbed too. Changing the left data or adaptive parameters needs additional terms.

Dissertation: Corollary 4.6.10.

How the reciprocal IRLS contraction bound is obtained

Write $\mathbf{B}_{\text{time}} = \operatorname{diag}(\mathbf{A}, \mathbf{A})$, $\mathbf{f} = \Phi(\mathbf{g})$, $\mathbf{f}' = \Phi(\mathbf{g}')$, and $\mathbf{r}' = \mathbf{y} - \mathbf{B}_{\text{time}}\mathbf{f}'$. Set $\mathbf{s} = \mathbf{B}_{\text{time}}\mathbf{g}$ and $\mathbf{s}' = \mathbf{B}_{\text{time}}\mathbf{g}'$. Subtract the frozen systems:

$$\mathbf{f} - \mathbf{f}' = \mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\left[\mathbf{W}^{-1/2}(\mathbf{W} - \mathbf{W}')\mathbf{r}'\right].$$

Use

$$\left\|\mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\right\| \le \ell^{-1/2}, \qquad \left\|\mathbf{W}'^{1/2}\mathbf{r}'\right\| \le \left\|\mathbf{W}'^{1/2}\mathbf{y}\right\|.$$

For reciprocal weights, $|w_i - w_i'| \le |s_i - s_i'|w_i w_i'$ and $w_i, w_i' \le \varepsilon^{-1}$.

Takeaway. Combining operator, residual, and weight-increment bounds gives $q = \|\mathbf{A}\|_2 \|\mathbf{y}\| / (\sqrt{\ell}\,\varepsilon^{3/2})$.

Dissertation: proof of Proposition 4.6.13; unclipped reciprocal map.

The frozen-weight fixed point and the likelihood stationary point differ

Consider the scalar Gaussian proxy with $y > 0$ and $\varepsilon > 0$, **without a regularizer**:

$$v(\theta) = |\theta| + \varepsilon, \qquad Q_k(\theta) = \frac{(y-\theta)^2}{2v(\theta_k)}.$$

Every frozen quadratic is minimized at $\theta = y$: the IRLS map has that fixed point. The state-dependent negative log-likelihood is

$$\mathcal{L}(\theta) = \frac{(y-\theta)^2}{2v(\theta)} + \frac{1}{2}\log v(\theta).$$

Yet $\mathcal{L}'(y) = 1/[2(y+\varepsilon)] > 0$.

Takeaway. The frozen-weight fixed point differs from the stationary point of this proxy likelihood.

Dissertation: Proposition 4.6.2. Scalar logical illustration outside the positive-Λ standing assumptions.

What the implemented IRLS procedure returns

At every candidate disparity:

1. Initialize from the unweighted exact solve.
2. Recompute predicted-rate weights, **clip for the solve**, and freeze them.
3. Solve the SPD system and damp toward its minimizer.
4. Recompute **unclipped scoring weights**; apply acceptance and stopping rules.
5. Return the last accepted iterate and its score.

The score omits the log-variance term. Masking, averaging, margin normalization, and SGM [95] then change candidate comparisons.

Takeaway. Brouwer and the contraction theorem describe fixed maps; early stopping and aggregation require their own analysis.

Dissertation: Section 4.5, Section 4.6.1; Section A.2.

Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth is finite and in the reporting depth band.
2. The returned estimate is finite.
3. The estimate also lies in the reporting band.

Configuration	Returned coverage	In-band retention	Scored points
Control	50.17%	51.54%	1.43 M
WLS	49.48%	96.40%	2.66 M

The scoring band is $[0.5, 8]$ m. Out-of-band finite depths are excluded by benchmark convention.

Dissertation: Table A.4, Weighting comparison (p. 182). Complete configurations; method-specific populations.

Exact zeros alone produce large pooled kurtosis

Suppose a residual distribution mixes a point mass at zero with a zero-mean Gaussian of variance $\sigma^2 > 0$, with zero-mass probability $0 \le \pi_0 < 1$.

$$\mathbb{E}X^2 = (1-\pi_0)\sigma^2, \qquad \mathbb{E}X^4 = 3(1-\pi_0)\sigma^4.$$

Its population excess kurtosis is

$$\frac{\mathbb{E}X^4}{(\mathbb{E}X^2)^2} - 3 = \frac{3}{1-\pi_0} - 3.$$

- Event-free pixels can have exactly zero fitted transport residual.
- Regularization, unequal variances, and temporal dependence affect diagnostics.
- Test event-active and variance-standardized fitted residuals before choosing a tail model.

Dissertation: Section 3.8.1; Model Mismatch in the Residuals.

Adaptive temporal weights are fixed in the theorem

For event activity $a_i = |dc_i|$, Chapter 2 uses these positive rules:

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}} \qquad \text{(sigmoid)}$$

$$\lambda_i = \max\left(10^{-3}, \min\left(\frac{100}{10^{-3} + a_i}, 100\right)\right) \qquad \text{(inverse proportional)}$$

$$\lambda_i = 10^{-3} + 10e^{-\min(a_i, 10)} \qquad \text{(exponential decay)}$$

Chapter 4 instead uses $\lambda_{xy} = \max\{\lambda_{\text{floor}}, \lambda_0 \|\mathbf{b}\| / (\|\mathbf{b}\| + 2)\}$.

Takeaway. Each fixed $\mathbf{\Lambda}$ is covered by the theorem; for the Chapter 2 rules, the data-dependent map $\mathbf{y} \mapsto \mathbf{\Lambda}(\mathbf{y}) \mapsto \mathbf{f}^*$ is locally Lipschitz (p. 98).

Dissertation: Adaptive temporal weights (p. 38); Section 4.5 (regularization terms); Theorem 2.2.5.

A real MVSEC flow example is qualitative

MVSEC indoor_flying2, frame 0340.



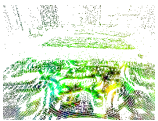
hue: direction
brightness: size, scaled within each panel (95th percentile); white: no flow value



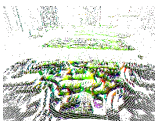
Reference



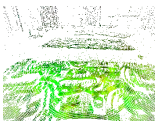
Ground truth



Event-accumulation baseline



Time surface



Voxel



Structure tensor voxel

Takeaway. One qualitative example; Chapter 3's AEE values use known-motion scenes.

Dissertation: Figure 3.2.

Quadratic H^1 regularization and TV answer different questions

The flow energy uses a quadratic gradient penalty:

$$E(\mathbf{v}) = \tfrac{1}{2}\int_\Omega \left(\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}\right) + \tfrac{\alpha^2}{2}\,\|\nabla \mathbf{v}\|_{L^2}^2\,, \qquad \mathbf{v} \in H^1(\Omega;\mathbb{R}^2).$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the stated hypotheses.
- TV [41] can preserve jumps, but needs a separate BV existence argument and its own uniqueness and stability assumptions.
- Normalized NE diffusion gives directional quadratic smoothing with a uniform ellipticity floor.

Takeaway. The integrated full-rank condition supports the quadratic H^1 proof; TV needs its own existence argument.

Dissertation: Theorem 3.6.1, function-space remark, Eq. 3.46.

Frozen weights match the value; an MM majorizer also matches the derivative

For a scalar prediction s, let $v(s) = |s| + \varepsilon$, $\varepsilon > 0$, and

$$S(s) = \frac{(y-s)^2}{2v(s)} + R(s), \qquad Q_k(s) = \frac{(y-s)^2}{2v(s_k)} + R(s).$$

The frozen surrogate matches S at s_k, but away from $s = 0$ its derivative omits

$$-\frac{v'(s)(y-s)^2}{2v(s)^2}.$$

Takeaway. A differentiable tangent majorizer must match the derivative at the current iterate; the frozen quadratic matches the value.

Dissertation: Section 4.6.1, Proposition 4.6.2; derivative consequence of the proposition.

A Gaussian model gives an evidence interpretation of attenuation

For common fixed $\mathbf{\Lambda} \succ 0$, $\gamma \geq 0$, and independent unit Gaussian noise, the candidate conditional is

$$\mathbf{y}_R \mid \mathbf{y}_L \sim \mathcal{N}(\mathbf{c}, (2\mathbf{P})^{-1}), \qquad \mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L.$$

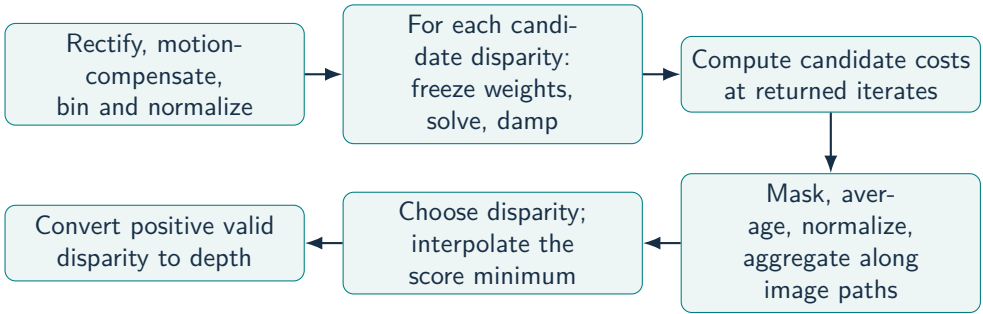
Coupled versus independent evidence satisfies

$$-\log \frac{p_\gamma(\mathbf{y}_L, \mathbf{y}_R)}{p_0(\mathbf{y}_L, \mathbf{y}_R)} = D(\mathbf{y}_L - \mathbf{y}_R) + \text{constant}.$$

Takeaway. The conditional mean explains attenuation; the discrepancy alone is an unevaluated alternative score.

Dissertation: Supplementary Gaussian derivation from Lemma 4.6.6, Proposition 4.6.8.

The implemented estimator has several distinct layers



Theorems describe specific layers; empirical evaluation describes the complete estimator.

Takeaway. The exact minimum of a computed score list certifies the best listed candidate for that score.

Dissertation: Section 4.5; Section A.2.

What does the dissertation contribute mathematically?

- A transparent temporal inversion with an explicit observation null space and quantitative fixed-weight guarantees.
- One well-posedness and stability theorem for several event-derived quadratic flow energies, with representation-specific hypotheses.
- A distinction between a coercive spline regularizer and a nonconvex contrast objective, plus full-objective existence.
- Stereo block structure, profiled-cost bias, ideal selection and robustness criteria, and a scoped adaptive-weight analysis.

Takeaway. Classical proof tools connect the models; the event-specific formulation, hypotheses, consequences, and limitations are the research content.

Dissertation: Section 5.3 and Table 5.1.

Literature anchors and where to find the claims

Image reconstruction

- Rebecq et al. [12]
  Events-to-Video.
- Zhang, Yezzi & Gallego [6]
  Image reconstruction as a linear inverse problem using optical flow.
- Antil–Sayre [2]
  Bilevel inverse problems in neuromorphic imaging.

Sensor overview

Gallego et al. [1]: *Event-Based Vision: A Survey.*

Flow models and analysis

- Horn–Schunck [16]
  Determining Optical Flow.
- Benosman et al. [50]
  Event-Based Visual Flow.
- Weickert–Schnörr [57]
  Convex regularizers for image motion.
- Gallego, Rebecq & Scaramuzza [3]
  A Unifying Contrast Maximization Framework.

Evaluation data

Zhu et al. [35]: *The Multivehicle Stereo Event Camera Dataset.*

Dissertation: Chapter 2; Section 3.2; Section A.2. Numbers match the dissertation bibliography; full entries on the closing reference pages.

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. Stability controls sensitivity to the data; accuracy is a separate property.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate; the physical sensor adds sub-threshold ambiguity.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Integrated directional information is necessary for uniqueness

Let Ω be connected, bounded, and Lipschitz; $\alpha > 0$; $\mathbf{B} \in L^\infty$ symmetric PSD; and $b \in L^2$. Minimize over unrestricted $H^1(\Omega; \mathbb{R}^2)$ fields. Set $M_B = \int_\Omega \mathbf{B}$. If M_B is singular, choose $c \neq 0$ with $M_B c = 0$. Then $\mathbf{B}c = 0$ a.e. and

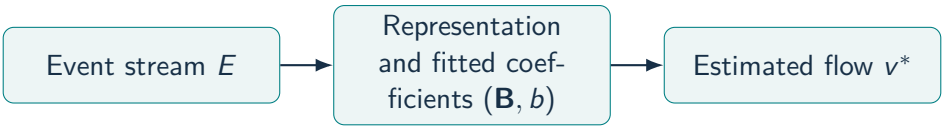
$$E(\mathbf{v} + tc) = E(\mathbf{v}) - t\, c^\top \int_\Omega b.$$

If the coefficient is nonzero, E is unbounded below; if zero, it is flat in direction c.

Takeaway. The integrated matrix condition is necessary for a unique minimizer in this unrestricted quadratic model.

Dissertation: Theorem 3.6.1; direct converse for its quadratic energy.

Coefficient stability is one stage of event-to-flow stability



The theorem controls coefficient-to-flow sensitivity. A Lipschitz representation map could compose with that bound.

Takeaway. An end-to-end guarantee also needs a metric and stability result for event preprocessing.

Dissertation: Theorem 3.6.1, Proposition 3.3.7; supplementary scope explanation.

Refinement requires a choice of continuum scaling

Illustrative smooth, uniform-time model only: spacing h, increments $y_i = hg_i$, and target energy

$$\frac{1}{2}\int |f' - g|^2 dt + \frac{\beta_{\text{time}}^2}{2}\int |f|^2 dt \quad \rightsquigarrow \quad \frac{1}{2h}\|\mathbf{A}f - y\|^2 + \frac{\beta_{\text{time}}^2 h}{2}\|f\|^2 .$$

Multiplying the discrete objective by h gives the Chapter 2 form with

$$\lambda_h = \beta_{\text{time}} h.$$

Takeaway. The penalty's grid scaling depends on the continuum objective being approximated.

Dissertation: Eq. 2.7, Eq. 2.8; supplementary smooth-model scaling illustration.

Strong smoothness selects one constant flow

For fixed $\Omega, \mathbf{B}, b$ satisfying the unified theorem, with unrestricted H^1 fields and the same α^2 gradient coefficient,

$$\alpha \to \infty \quad \Longrightarrow \quad v_\alpha \to c \text{ in } H^1, \qquad c = \left(\int_\Omega \mathbf{B} \right)^{-1} \int_\Omega b.$$

Takeaway. Very strong spatial regularization removes variation and leaves the data-balanced constant field.

Dissertation: Theorem 3.6.1; direct strong-regularization limit.

Transporting a scalar differs from conserving a density

Scalar transport and source-free density conservation are

$$V_t + v \cdot \nabla V = 0, \qquad V_t + \nabla \cdot (Vv) = 0, \qquad \nabla \cdot (Vv) = v \cdot \nabla V + V \nabla \cdot v.$$

They coincide when $V \nabla \cdot v = 0$, in particular for divergence-free flow.

Takeaway. The continuity equation adds a density-compression term to scalar transport.

Dissertation: Section 3.4.1, Scalar transport and density conservation (p. 85); direct product-rule comparison.

Correlated errors change the uncertainty; coercivity holds

For a centered symmetric regression design Z with $Q = Z^\top Z \succ 0$, timestamp-error covariance Σ gives

$$\mathrm{Cov}(\hat{g}) = Q^{-1} Z^\top \Sigma Z Q^{-1}.$$

This reduces to $\sigma^2 Q^{-1}$ when $\Sigma = \sigma^2 I$.

With any $\Gamma \succ 0$, the temporal system matrix is $\mathbf{A}^\top \Gamma^{-1} \mathbf{A} + \mathbf{\Lambda}^2 \succ 0$ when $\mathbf{\Lambda} \succ 0$.

Takeaway. The deterministic quadratic solve stays coercive, while statistical uncertainty depends on the error covariance.

Dissertation: Proposition 3.3.7; Eq. 2.8, Theorem 2.2.5; direct correlated-error extension.

Optical flow: further analysis

- Time-surface gradient estimation: bias, variance, and design.
- Contrast maximization: what it models, known-motion evidence, objective, and existence proof.

Regression quantifies coefficient uncertainty; contrast maximization changes the objective being optimized.

Time-surface gradient estimation: bias versus variance

Fit a plane to timestamps at offsets $\mathbf{z}_j$ around a pixel.

$$Q = \sum_j \mathbf{z}_j \mathbf{z}_j^\top \succ 0.$$

For a centrally symmetric design and independent timestamp errors with common mean and finite variance σ^2:

$$\mathrm{Cov}(\widehat{\nabla T}) = \sigma^2 Q^{-1}.$$

With a Lipschitz Hessian of constant L_H on a neighborhood containing the sample disk,

$$\|\mathrm{Bias}\| \le \frac{L_H}{6} \left\|Q^{-1}\right\| \sum_j \|\mathbf{z}_j\|^4, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \sigma^2 \,\mathrm{tr}(Q^{-1}).$$

Takeaway. A larger neighborhood reduces variance but can increase curvature bias. Regression rank and velocity identifiability are different questions.

Dissertation: Proposition 3.3.7.

Contrast maximization finds the motion that sharpens the events

What it models

- A moving edge leaves a smeared trail of events over the time window.
- Slide each event back along a candidate flow to one reference time. The correct flow stacks the trail into a sharp edge; image variance measures the sharpness.
- It uses individual events in continuous time, with no time surface, bins, or fitted gradients. Binned models blur when an edge moves during a bin.

Known-motion evidence

AEE (px) at window 22; a zero flow field scores 3.00.

Scene	Best other (struct. tensor)	CMAX
Idealized stripes	2.15	**0.49**
Simulated stripes (v2e)	1.92	**1.19**
Simulated checkerboard (v2e)	0.98	**0.32**

On the stripes, CMAX recovers the diagonal motion: warping the events over the whole window brings in evidence from the object's boundary.

Takeaway. CMAX has the lowest error in every known-motion scene.

Dissertation: Section 3.5; Figure 3.3 and Figure 3.4; contrast maximization [3, 55]; v2e [46].

How the known-motion comparison was evaluated

Aspect	Reported protocol
Stimulus	A 60×40 px object translates diagonally at 3 px per window; direct idealized events and v2e-simulated events [46].
Settings	Quadratic variants: $\alpha_{\text{smooth}} = 20$, no pyramid. Contrast maximization: 100 L-BFGS [71] iterations.
Selection	Window 22 minimizes summed deviation from each variant's median AEE over windows 1–31.
Display / error	Common arrow scale; AEE is mean Euclidean displacement error, in pixels per evaluation window.

Equal numerical α values give different stencil-dependent smoothing strengths.

Takeaway. These controlled examples show directional-information effects at the reported solver settings.

Dissertation: Figure 3.3 and Figure 3.4; smoothing stencils (p. 164).

Contrast maximization changes the optimization problem

Warp each event to a common normalized reference time:

$$\mathbf{x}_k^w = \mathbf{x}_k + (\widetilde{t}_{\mathrm{ref}} - \widetilde{t}_k)\mathbf{q}(\mathbf{x}_k), \qquad \mathbf{q} = T\mathbf{v}_{\mathrm{phys}}.$$

Represent $\mathbf{q}$ by a tensor-product cubic B-spline on a uniform knot grid ($h = 8\,\mathrm{px}$), with coefficients $\mathbf{c}$; both components share the basis.

$$F(\mathbf{c}) = -\operatorname{Var}(I_{\mathbf{c}}) + \mathbf{c}^\top \mathbf{K}_{\mathrm{reg}}\mathbf{c}, \qquad \mathbf{K}_{\mathrm{reg}} \succeq \tau_{\mathrm{ridge}}\mathbf{I}, \quad \tau_{\mathrm{ridge}} > 0.$$

- A good warp concentrates aligned events.
- A flexible incorrect warp can also concentrate events by collapsing them [55].

Takeaway. The full objective is generally nonconvex, outside the quadratic flow theorem.

Dissertation: Section 3.5, Proposition 3.5.1, Proposition 3.5.2; contrast maximization [3].

A global minimizer exists, even though the solver is local

For finitely many fixed event weights, let $W_{\mathrm{tot}} = \sum_k |w_k|$. The bilinear accumulation is continuous and satisfies

$$\mathrm{Var}(I_{\mathbf{c}}) \le \frac{W_{\mathrm{tot}}^2}{HW}.$$

Therefore

$$F(\mathbf{c}) \ge -\frac{W_{\mathrm{tot}}^2}{HW} + \tau_{\mathrm{ridge}} \|\mathbf{c}\|^2 \longrightarrow \infty.$$

Proof mechanism

Continuous objective + bounded focus gain + coercive ridge $\Rightarrow$ compact sublevel set $\Rightarrow$ Weierstrass.

Takeaway. A global minimizer exists by finite-dimensional coercivity; the local solver returns a local solution.

Dissertation: Proposition 3.5.2 and evaluated-solver scope.

Stereo depth: full Chapter 4 arc

1. Coupled model and the frozen-weight inner solve.
2. Sum and difference modes; the profiled candidate cost.
3. The explicit counterexample and the correct-selection theorem.
4. Winning-margin stability and the non-identifiability example.
5. Heteroscedastic weights and IRLS fixed-point analysis.
6. MVSEC benchmark, weighting comparison, and hard fusion.

These results separate the inner temporal solve from the outer correspondence decision.

Stereo depth in one slide: what Chapter 4 adds

At each candidate disparity d, solve two coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

1. **Inner solve.** The Hessian is $\succeq \lambda_{\min}^2 \mathbf{I}$: the temporal proof plus a nonnegative coupling term.
2. **Profiled cost.** $C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + D_d$: the monocular term moves with d. In an exact two-sample example the perfect match costs $\frac{1}{3}$, an attenuated one $\frac{9}{40}$: a weaker right-view copy can outscore the true match.
3. **Selection.** A separation condition makes the correct candidate the unique winner; a margin Δ tolerates cost perturbations below $\Delta/2$ (e.g., noise events, threshold mismatch between sensors).
4. **Reweighting.** The IRLS map has a fixed point (Brouwer) and contracts under a sufficient condition.

MVSEC indoor flying: soft-coupled WLS is best on Flying 2 (58.35 cm) and log RMSE $\times 100$ (44.54), second elsewhere, with 2.66 M scored points versus 1.20 M for the strongest prior method.

Takeaway. Coupling keeps every inner solve well posed; the right correspondence also needs a cost margin.

Dissertation: Theorem 4.6.3, Proposition 4.6.7, Proposition 4.6.8, Proposition 4.6.13; Table A.5; MVSEC [35], Piatkowska [78], Brouwer [92].

Stereo alignment after motion compensation

The model pairs synchronized, rectified observations of a scene point visible in both cameras.

Each camera fits its own translation within each temporal slice

$$\mathbf{x}'_c = \mathbf{x}_c + (t_{\mathrm{ref}} - s)\mathbf{v}_c, \qquad c \in \{L, R\}.$$

For simultaneous rectified correspondences,

$$d_{\mathrm{w}}(s) = d(s) + (t_{\mathrm{ref}} - s)(v_{x,L} - v_{x,R}).$$

- A horizontal warp mismatch changes the disparity presented to the matcher.
- A vertical warp mismatch moves a correspondence away from its scanline.
- Reported slices are 12.5 ms wide within a 0.25 s evaluation window.

Takeaway. A unique inner solve fits the supplied observations; physical correspondence also relies on compatible timing, calibration, and warps.

Dissertation: Motion-compensated accumulation (p. 108); Table A.1.

At each candidate, couple two temporal profiles

Suppress the pixel index; $\mathbf{y}_R = \mathbf{y}_R(d)$.

$$J_d(\mathbf{f}_L, \mathbf{f}_R; \mathbf{W}_L, \mathbf{W}_R) = \frac{1}{2}\left\|\mathbf{W}_L^{1/2}(\mathbf{A}\mathbf{f}_L - \mathbf{y}_L)\right\|^2 + \frac{1}{2}\left\|\mathbf{W}_R^{1/2}(\mathbf{A}\mathbf{f}_R - \mathbf{y}_R)\right\|^2 + \frac{1}{2}\|\mathbf{\Lambda}\mathbf{f}_L\|^2 + \frac{1}{2}\|\mathbf{\Lambda}\mathbf{f}_R\|^2 + \frac{\gamma}{2}\|\mathbf{f}_L - \mathbf{f}_R\|^2 .$$

- Data fidelity estimates each temporal profile from its own view.
- Magnitude penalties control temporal baselines.
- Soft coupling penalizes left–right disagreement.

Takeaway. The right latent profile is auxiliary: one per pixel and candidate.

Dissertation: Eq. 4.24; fixed-disparity model definition.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Sum and difference modes expose the coupling

For equal blocks $\mathbf{K}_L = \mathbf{K}_R = \mathbf{K}$, introduce

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}, \qquad \mathbf{U}^\top \mathbf{M} \mathbf{U} = \begin{pmatrix} \mathbf{K} & 0 \\ 0 & \mathbf{K} + 2\gamma \mathbf{I} \end{pmatrix}.$$

Mean profile
The coupling cancels. Shared information is fused by the temporal solve.

Difference profile
The coupling raises curvature and suppresses disagreement.

For $\mathbf{\Lambda} = \lambda \mathbf{I}$ in the unweighted model,

$$\kappa_2(\mathbf{M}) = 1 + \frac{g_{\max} + 2\gamma}{\lambda^2}.$$

Dissertation: Lemma 4.6.15, Theorem 4.6.16.

Profile out the latent variables to see what disparity compares

In the unweighted model, let $\mathbf{K} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}$ and

$$m(\mathbf{y}) = \frac{1}{2} \mathbf{y}^\top (\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top)\mathbf{y}.$$

The exact minimized candidate cost is

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + \frac{1}{4}\boldsymbol{\delta}_d^\top \left[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}\right] \boldsymbol{\delta}_d, \quad \boldsymbol{\delta}_d = \mathbf{A}^\top(\mathbf{y}_L - \mathbf{y}_R(d)).$$

- The final term measures left–right discrepancy.
- The right monocular cost also changes with the candidate.

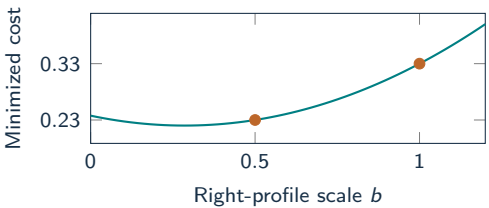
Takeaway. The minimized regularized energy is more than a correspondence discrepancy.

Dissertation: Lemma 4.6.6, Proposition 4.5.1. Equal fixed weights admit a scaled analogue.

An attenuated match can cost less than the perfect one

Take $\mathbf{A} = [-1, 1]$, $\mathbf{\Lambda} = \mathbf{I}$, $\gamma = 1$, $y_L = 1$, and $y_R = b$.

$$C(b) = \frac{1 + b^2}{6} + \frac{(1 - b)^2}{15}.$$



Let D denote the discrepancy term in the preceding profiled-cost formula.

$$C(1) = \frac{1}{3}, \qquad C\left(\frac{1}{2}\right) = \frac{9}{40} < \frac{1}{3}.$$

$$b^* = \frac{D(y_L)}{m(y_L) + D(y_L)} = \frac{2}{7}.$$

An attenuated wrong candidate beats the exact match if both are available.

Takeaway. Inner uniqueness, distinct observations, and correct correspondence are separate claims.

Dissertation: Proposition 4.6.7, Correspondence identifiability (p. 143). Unweighted cost before aggregation.

Correct disparity needs a separation condition

Write the unweighted cost as

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_d) + D(\mathbf{y}_L - \mathbf{y}_d), \qquad D(\mathbf{e}) \geq 0.$$

A sufficient correct-selection theorem

On a finite candidate set, suppose $\mathbf{y}_{d_*} = \mathbf{y}_L$, the positive regularizer is common to both views and all candidates, and $\gamma > 0$ is candidate-independent. If every competitor has

$$\mathbf{y}_d \neq \mathbf{y}_L \quad \text{and} \quad m(\mathbf{y}_d) \geq m(\mathbf{y}_L),$$

then $C(d) > C(d_*)$: the correct candidate is the unique winner.

Takeaway. The discrepancy must overcome candidate-dependent monocular bias. Stronger metric-separation criteria are also available.

Dissertation: Proposition 4.6.8(b1); additional criterion in backup.

A winning margin turns cost stability into selection stability

For a fixed finite set of at least two candidates with a unique winner d_*, define

$$\Delta = \min_{d \neq d_*} \left(C(d) - C(d_*) \right) > 0.$$

If $|\widetilde{C}(d) - C(d)| \leq \eta$ for all candidates, including d_*, then every $d \neq d_*$ satisfies

$$\widetilde{C}(d) - \widetilde{C}(d_*) \geq \Delta - 2\eta > 0 \qquad \text{when } \eta < \Delta/2.$$

A non-identifiability example

Two candidates with identical right-view profiles have identical local costs when the candidate procedures and parameters coincide.

Takeaway. Correspondence separation depends on the profiles' coupling and attenuation as well as temporal activity.

Dissertation: Corollary 4.6.10, Proposition 4.6.20.

Depth sensitivity grows with distance

For fixed calibration and positive d and $d + \delta d$,

$$Z(d) = \frac{fB}{d}, \qquad |Z(d + \delta d) - Z(d)| = \frac{fB|\delta d|}{d(d + \delta d)}.$$

Small relative disparity error

$$|\delta Z| \approx \frac{Z^2}{fB}|\delta d|, \qquad \frac{|\delta Z|}{Z} \approx \frac{|\delta d|}{d}.$$

At the same disparity precision, absolute depth error grows quadratically with distance.

Illustrative exact calculation

	Reference d	Estimated d
Farther point	4 px	5 px
Nearer point	20 px	21 px

Depth is underestimated by 20% and 4.76%, respectively.

Takeaway. Read depth errors with the disparity precision, depth range, calibration, and scored population.

Dissertation: Eq. 4.11 and Eq. 4.12; stereo geometry [80]; example percentages are illustrative arithmetic.

Heteroscedastic weights: statistical motivation and scope

As a working raw-count model,

$$N_+ \sim \text{Poisson}(\theta_+), \qquad N_- \sim \text{Poisson}(\theta_-),$$

independently. For $b = N_+ - N_-$ (Skellam [87]),

$$\mathbb{E}b = \theta_+ - \theta_-, \qquad \text{Var}(b) = \theta_+ + \theta_-.$$

$$|\theta_+ - \theta_-| \leq \theta_+ + \theta_-, \quad \text{with equality iff at least one rate is zero.}$$

The evaluated reciprocal proxy is

$$W_{ii}^{(k)} = \frac{1}{|(\mathbf{A}\mathbf{f}^{(k)})_i| + \varepsilon_{\text{noise}}}, \qquad \varepsilon_{\text{noise}} > 0.$$

Takeaway. Total variance follows the sum of the two event rates. The processed-data weights define a frozen Gaussian surrogate.

Dissertation: Definition 4.6.1 and following scope discussion.

IRLS: fixed-point existence and conditional convergence

Fix d, $\mathbf{\Lambda}$, and γ; stack $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)$. Let $\Phi(\mathbf{g})$ minimize the quadratic with weights frozen at $\mathbf{g}$. For the *unclipped reciprocal map*, set $\ell = \lambda_{\min}^2 > 0$:

$$\|\Phi(\mathbf{g})\|^2 \leq \frac{\|\mathbf{y}\|^2}{\ell\, \varepsilon_{\text{noise}}}.$$

A sufficiently large closed ball is invariant; continuity and Brouwer [92] give a fixed point.

Sufficient convergence condition

$$q = \frac{\|\mathbf{A}\|_2 \, \|\mathbf{y}\|}{\sqrt{\ell}\, \varepsilon_{\text{noise}}^{3/2}} < 1 \quad \Rightarrow \quad \text{unique fixed point and geometric convergence.}$$

For fixed damping $\omega \in (0, 1]$, the bound is $q_\omega = 1 - \omega + \omega q$.

Takeaway. A fixed point exists; convergence follows under the displayed sufficient condition.

Dissertation: Lemma 4.6.12, Proposition 4.6.13, Section 4.6.1; IRLS [84].

Stereo benchmark: read accuracy alongside scored coverage

Method	Mean depth error (cm) ↓ Fl1	Fl2	Fl3	Fl4	log RMSE ×100 ↓	Scored points (M)
Unweighted control	140.23	115.40	142.35	113.00	96.18	1.43
Unweighted SGM [95]	83.45	72.90	85.33	71.62	50.55	2.39
Soft-coupled WLS	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20
Xie adaptation [105]	93.43	78.60	93.66	66.57	59.36	1.39
Kogler et al. [104]	123.39	99.50	125.27	78.23	80.94	0.66
Rogister et al. [103]	142.17	128.20	153.32	114.80	90.32	1.01

MVSEC [35] indoor flying, raw outputs. Each method is scored on its own estimates and ground truth in $[0.5, 8]$ m; points summed and log RMSE averaged over sequences.

Takeaway. WLS is competitive with a larger scored population.

Dissertation: Table A.5, Table A.4; Xie and Piatkowska rows are local adaptations.

A weighting comparison supports the selected rule

Compare reciprocal versus rate-proportional weights while holding the remaining configuration fixed.

	Flying 1	Flying 2	Flying 3	Flying 4
Mean-error difference (cm)	4.44	8.28	2.73	5.17

The reciprocal rule has lower reported mean error on all four sequences.

Scope of this evidence

Each method is scored on its own valid pixel set. The differences subtract the reported WLS means from the WLS-Prop means; the rule was selected on these sequences.

Takeaway. The reported comparison supports the chosen rule on its scored populations; a common-set comparison requires rescoring the outputs.

Dissertation: Table A.5, Weighting comparison (p. 182).

Hard fusion changes both the constraint and the regularization

Joint intensity fusion (JIF) imposes $\mathbf{f}_L = \mathbf{f}_R = \mathbf{v}$ and solves

$$(2\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)\mathbf{v} = \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

For fixed equal weights, the unchanged soft model has the limit

$$\gamma \to \infty : \qquad \mathbf{f}_L, \mathbf{f}_R \to (\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)^{-1} \frac{1}{2} \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

JIF instead uses $\mathbf{A}^\top \mathbf{W}\mathbf{A} + \frac{1}{2}\Lambda^2$ after dividing its equation by two.

Takeaway. Reported mean depth errors: JIF 102.10 cm; soft WLS 65.44 cm. Hard equality, regularization, weights, iteration budgets, and scored populations differ.

Dissertation: Proposition 4.9.1, Section 4.9.3.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

One complementarity lemma behind the uniqueness results

Lemma. On a Hilbert space H, let $a, r \geq 0$ be bounded symmetric forms and $N \subset H$ finite-dimensional with orthogonal projection P. If

$$r(u,u) \geq \rho \|u - Pu\|^2, \qquad a(n,n) \geq \mu \|n\|^2 \quad (n \in N), \qquad a(u,u) \leq \beta \|u\|^2,$$

then $a(u,u) + r(u,u) \geq c_0 \|u\|^2$ with $c_0 = \min\{\mu/2,\ \rho\mu/(\mu + 2\beta)\}$. When $N = \ker r$, $\mu > 0$ is also necessary.

Problem	r controls all but N	N	a must see N
Temporal (Ch. 2)	data $\|\mathbf{Af}\|^2$, spectral gap g_1	constants	penalty: $\sum_i \lambda_i^2 > 0$
Flow (Ch. 3)	$\alpha^2 \|\nabla \mathbf{v}\|_{L^2}^2$, Poincaré	constant fields	data: $\lambda_{\min}(\int_\Omega \mathbf{B}) > 0$
Discrete flow	connected graph Laplacian	constant fields	data: $\sum_i q_i B_i \succ 0$
Stereo inner (Ch. 4)	data $+\gamma \|\mathbf{f}_L - \mathbf{f}_R\|^2$; $\mathbf{W}_c \succ 0$, $\gamma > 0$	common constants	penalty: $\mathbf{\Lambda} \neq 0$

Takeaway. One term controls all but a finite-dimensional mode; the other must see it. The roles swap between temporal and flow, and c_0 falls linearly with μ.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 3.7.1, Theorem 4.6.3. The common lemma restates their proofs.

Adaptive weights keep the temporal estimate locally Lipschitz

Let $\lambda_i = \varphi(|dc_i|)$ with $0 < \lambda_- \le \varphi \le \lambda_+$ and φ L-Lipschitz. Subtracting the normal equations,

$$\mathbf{K}_1(\mathbf{f}_1^* - \mathbf{f}_2^*) = \mathbf{A}^\top(\mathbf{y}_1 - \mathbf{y}_2) - (\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2)\mathbf{f}_2^*.$$

With $\left\|\mathbf{K}^{-1}\mathbf{A}^\top\right\| \le \frac{1}{2\lambda_-}$, $\left\|\mathbf{K}^{-1}\right\| \le \lambda_-^{-2}$, $\left\|\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2\right\| \le 2\lambda_+ L \left\|dc_1 - dc_2\right\|_\infty$:

$$\|\mathbf{f}_1^* - \mathbf{f}_2^*\| \le \frac{\|\mathbf{y}_1 - \mathbf{y}_2\|}{2\lambda_-} + \frac{\lambda_+ L}{\lambda_-^3}\, \|\mathbf{y}_2\|\, \left\|dc_1 - dc_2\right\|_\infty .$$

Chapter 2 rule	λ_-	λ_+	L
Sigmoid	10^{-3}	5.001	2.5
Inverse proportional	10^{-3}	100	100
Exponential decay	$10^{-3} + 10e^{-10}$	10.001	10

Takeaway. Each Chapter 2 rule gives a locally Lipschitz estimator with explicit, pessimistic constants; Chapter 4's solution-dependent IRLS weights need the fixed-point analysis.

Dissertation: Theorem 2.2.5; adaptive rules (p. 38). The Lipschitz bound is a short argument beyond the chapter.

How the parameters were chosen

Problem	Parameters	Selection
Temporal (Ch. 2)	Scale of each $\mathbf{\Lambda}$ rule; kernel and width	Grid sweep per rule; three rules compared by fidelity and CN
Flow (Ch. 3)	Smoothness weight α	Grid sweeps on MVSEC
Stereo (Ch. 4)	Coupling γ, latent count; weighting	Grid sweep; paired weighting comparison on the four sequences

Antil–Sayre [2]: scalar weight λ_2. Chapter 2: adaptive diagonal $\mathbf{\Lambda}$, three rules evaluated.

What the theorems guarantee

The results hold for every positive value: sweeps tune bias and accuracy; uniqueness and stability hold throughout.

Principled selection rules

Discrepancy principle (Morozov 1966; Skellam noise level $\theta_+ + \theta_-$), L-curve (Hansen 1992), GCV (Golub–Heath–Wahba 1979), bilevel learning.

Takeaway. Sweeps chose the values; the theorems cover every positive choice.

Dissertation: Adaptive rules (p. 38); Table 2.7; flow sweeps (p. 164); Section 4.8; Definition 4.6.1. Parameter-choice rules: external works named above.

Learned estimators face the same free directions

Problem	Left free by the events	Model-based choice	Learned choice
Temporal	Constant offset at each pixel (Lemma 2.2.4)	Magnitude penalty $\mathbf{\Lambda}$ (Theorem 2.2.5)	Brightness learned from training images [12]
Flow	Motion along a straight edge (Figure 3.1)	H^1 smoothness, integrated data (Theorem 3.6.1)	Spatial context learned from data [39]
Stereo	Candidates with coincident right-view profiles (Proposition 4.6.20)	Spatial cost aggregation (SGM)	Learned matching cost

Combining them

Learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{\Lambda} \mapsto \mathbf{f}^*(\mathbf{\Lambda})$ is single-valued and smooth.

Data without a training set

ISS and ballistics: reconstructed from events alone, without reference images or training data.

Takeaway. The theorems separate what the events determine from what the regularizer supplies, in either form.

Dissertation: Lemma 2.2.4, Theorem 2.2.5, Theorem 3.6.1, Proposition 4.6.20; qualitative demonstrations (p. 44). External paper cited above.

The spline regularizer is uniformly coercive under refinement

Cubic B-splines, uniform knot spacing h: $\Phi_j \geq 0$, $\sum_j \Phi_j \equiv 1$, $\int_\Omega \Phi_j \leq h^2$, and $u_h = \sum_j c_j \Phi_j$.

Step 1: coefficients control the L^2 norm

$u_h^2 \leq \sum_j c_j^2 \Phi_j$ pointwise, so $\|u_h\|_{L^2}^2 \leq h^2 \|\mathbf{c}\|^2$.

Step 2: both regularizer terms, for each flow component

$$\beta\, \mathbf{c}^\top \mathbf{K}_0 \mathbf{c} + \tau_{\text{ridge}} \|\mathbf{c}\|^2 \;\geq\; \beta \varepsilon_{\text{ell}} \|\nabla u_h\|_{L^2}^2 + \frac{2\varepsilon_{\text{ridge}}}{Mh^2} \|u_h\|_{L^2}^2 .$$

$Mh^2 \to |\Omega|$ (ratio 1.20 at $h = 8\,\text{px}$), so the H^1 coercivity constant stays bounded below as $h \to 0$; $\kappa_2(\mathbf{K}_{\text{reg}}) = O(h^{-2})$, the usual stiffness-plus-mass growth.

Takeaway. The ridge eigenvalue shrinks with refinement; the function-space penalty stays uniform.

Dissertation: Proposition 3.5.1; contrast-maximization parameters (p. 166). The uniform bound is a short argument beyond the chapter.

Weak regularization: a limit in time, the aperture line in space

Temporal, $\mathbf{\Lambda} = \lambda \mathbf{D}$, fixed diagonal $\mathbf{D} \succ 0$

- Every $\lambda > 0$: $\mathbf{1}^\top \mathbf{D}^2 \mathbf{f}_\lambda = 0$, so the baseline rule is independent of λ.
- $\lambda \to 0$: $\mathbf{f}_\lambda \to \operatorname{argmin}\{\|\mathbf{Du}\| : \mathbf{Au} = \mathbf{y}\}$, an exact fit of the increments.
- $\mathbf{D} = \mathbf{I}$: noise gain $\left\|\mathbf{K}_\lambda^{-1}\mathbf{A}^\top\right\| \leq \min\{\frac{1}{2\lambda}, \frac{1}{\sqrt{g_1}}\}$, with $\frac{1}{\sqrt{g_1}} \approx \frac{N_t}{\pi}$.

Flow, smoothness weight α^2

- If $\mathbf{B} = 0$ on an open $U \subset \Omega$: for $\mathbf{v} \in H_0^1(U; \mathbb{R}^2)$, $\mathcal{B}(\mathbf{v}, \mathbf{v}) = \alpha^2 \|\nabla \mathbf{v}\|^2 \leq \alpha^2 \|\mathbf{v}\|_{H^1}^2$, so the coercivity constant is at most α^2.
- $\alpha = 0$, time surface: the minimizers are all $\mathbf{v}$ with $\nabla \mathcal{T} \cdot \mathbf{v} = 1$ wherever $\nabla \mathcal{T} \neq 0$, the aperture line at each pixel.

Takeaway. Without the temporal penalty the increments are fit exactly and noise accumulates along the chain; without spatial smoothing each pixel returns to its aperture line.

Dissertation: Regularized formulation (p. 16); Lemma 4.4.1; Proposition 4.6.23; Theorem 3.6.1; Proposition 3.3.4. The flow statements are a short argument beyond the chapter.

Where there are no events, smoothness fills in harmonically

Let $U \subset \Omega$ be open, connected, and Lipschitz, with $\mathbf{B} = 0$ and $b = 0$ a.e. in U, and $\Gamma = \partial U \cap \Omega \neq \emptyset$.

The weak equation inside U

For every $\boldsymbol{\eta} \in H^1_0(U; \mathbb{R}^2)$: $\int_U \nabla \mathbf{v}^* : \nabla \boldsymbol{\eta} = 0$. Each component of $\mathbf{v}^*$ is harmonic, hence smooth, in U; on U it minimizes $\int_U |\nabla \mathbf{w}|^2$ given its values on Γ.

Maximum principle

For a unit vector $\mathbf{e}$: if $\mathbf{e} \cdot \mathbf{v}^* \leq M$ on Γ, test with $\boldsymbol{\eta} = (\mathbf{e} \cdot \mathbf{v}^* - M)^+ \mathbf{e}$ on U, zero elsewhere. Then $\alpha^2 \int_U |\nabla \boldsymbol{\eta}|^2 = 0$, so $\mathbf{e} \cdot \mathbf{v}^* \leq M$ in U. Over all $\mathbf{e}$: $\sup_U |\mathbf{v}^*| \leq \sup_\Gamma |\mathbf{v}^*|$.

Takeaway. An event-free region receives the least-gradient interpolation of the motion measured around it, with no speed larger than on its boundary.

Dissertation: Theorem 3.3.9; Theorem 3.6.1. Harmonicity and the maximum principle are a short argument beyond the chapter.

What can changes tell us about a scene?



Reference image

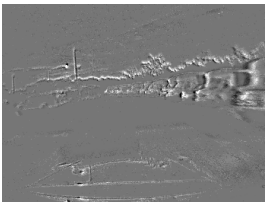


Accumulated events

50 ms sample

Orange +; blue −; purple: both.

→



Our reconstruction

What do the observations determine, and what does the model supply?

Takeaway. The observations constrain changes; the model supplies the remaining choices.

Dissertation: Chapter 2, saved MVSEC [35] outdoor_day1 reconstruction example.

Where this work fits

Image reconstruction

Rebecq et al. [12]: recurrent network, events to video.

Zhang, Yezzi & Gallego [6]: optical flow supplies a linear image inverse problem; classical or learned priors.

Antil–Sayre [2]: events and frames; bilevel estimation of an event-based blur-kernel parameter.

This work: per-pixel temporal inversion from event increments; kernel representations, activity-based penalties, and fixed-weight guarantees.

Optical flow

Horn–Schunck [16]: brightness constraint and spatial smoothness.

Benosman et al. [50]: timestamp geometry.

Weickert–Schnörr [57]: convex-regularizer framework and well-posedness theory.

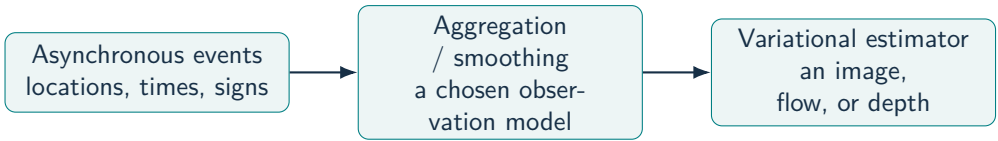
This work: three event representations in one quadratic theory, with explicit identifiability and stability assumptions.

Takeaway. The contribution is the event-derived models and their precise guarantees: what the data determine and what regularization supplies.

Dissertation: Chapter 2 and Section 3.2; temporal publication: Antil, Blauvelt, and Sayre [17].

A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$



- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

Takeaway. The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Eq. 2.2–Eq. 2.4; Section 3.3; Section 4.3.3.

Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

One spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$: the bound limits how far one noise event moves a pixel's history.

Takeaway. Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Theorem 2.2.5, Proposition 2.2.7.

How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here dc_i is the signed event accumulation; a_i is its magnitude.

Rule	Sigmoid	Inverse proportional	Exponential
Reported outdoor CN	29.60	81.50	100.74

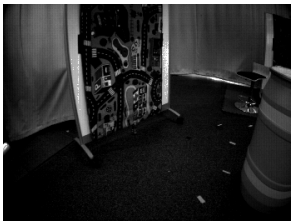
Selected by grid sweeps, balancing image fidelity and conditioning.
Quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.

Takeaway. Positive weights ensure a unique solve; the stability bound holds with weights fixed.

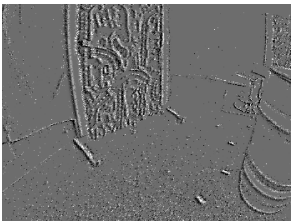
Dissertation: Adaptive temporal weights (p. 38); Table 2.7.

Temporal evidence: events fix structure; the penalty sets the baseline

MVSEC [35] indoor flying 2



Reference



Event-only reconstruction

MVSEC indoor aggregate results

Kernel	PSNR	CN
B-spline	8.86	22.16
Delta	7.70	1375.75
Gaussian	8.84	20.22

PSNR compares with a reference frame, so it also scores the absolute brightness that the model selects; the events constrain only changes.

Takeaway. Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Table 2.4; images: MVSEC indoor flying 2, Ch. 2.

Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

In scene terms: $|\nabla\mathcal{T}|$ is edge slowness, normal speed $1/|\nabla\mathcal{T}|$; motion along the edge leaves $\mathcal{T}$ fixed.

Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

Takeaway. Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Proposition 3.3.4; Definition 3.3.2; time surfaces [49, 50].

Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$; project over a window, $a(\mathbf{x}, s) = \int_s^{s+T} V(\mathbf{x}, t)\, dt$.

Modeling assumption (added to the threshold model)

The activity field moves with the scene: $V_t + \mathbf{v} \cdot \nabla V = 0$, with constant $\mathbf{v}$ over the window.

Integrating over the window then gives

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s),$$

so the voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

Takeaway. The threshold model supplies the activity field; transporting it with the scene is a separate modeling choice.

Dissertation: Section 3.4.1; derivative-operator qualifications; voxel grid [39].

The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

Takeaway. Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Section 3.4.2, Proposition 3.4.4; spatiotemporal-surface view [49].

Proof idea: smoothness fixes variation, edges fix the drift

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

Smoothness controls variation

$$\|\mathbf{w}\|_{L^2} \le C_P \|\nabla \mathbf{v}\|_{L^2}$$

Poincaré [26]

Data control constant motion

$$m = \lambda_{\min}\Big(\int_\Omega \mathbf{B}\, d\mathbf{x}\Big) > 0$$

edges cover every direction

Together: coercivity

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) \ge c_0 \|\mathbf{v}\|_{H^1}^2$$

$$c_0 = c_0(\Omega, \alpha, \|\mathbf{B}\|_{L^\infty}, m)$$

In scene terms: smoothness leaves a camera pan free, so the edges must fix it; c_0 shrinks with m, as for nearly parallel stripes (next slide).

Takeaway. The gradient penalty controls nonconstant variation; integrated directional information controls the constant modes. Lax–Milgram [72] then gives the unique minimizer.

Dissertation: proof of Theorem 3.6.1.

Stereo on MVSEC: competitive accuracy, more scored points

At each candidate disparity d, solve two softly coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2}(\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda}\mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

Mathematical result (fixed d, frozen positive weights): Hessian $\succeq \lambda_{\min}^2 \mathbf{I}$; unique inner solve.
Benchmark: the full WLS method, with weight updates, spatial cost aggregation, and candidate selection.

Method	Mean depth error (cm) ↓ Fl1	Fl2	Fl3	Fl4	log RMSE ×100 ↓	Scored points (M)
Soft-coupled WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20
Xie adaptation [105]	93.43	78.60	93.66	66.57	59.36	1.39
Kogler et al. [104]	123.39	99.50	125.27	78.23	80.94	0.66
Rogister et al. [103]	142.17	128.20	153.32	114.80	90.32	1.01

MVSEC indoor flying 1–4, raw outputs. Each method is scored on its own estimates with ground truth in $[0.5, 8]$ m.

Takeaway. WLS: best on Flying 2 and log RMSE, within 6 cm elsewhere, over twice the points.

Dissertation: Theorem 4.6.3; Table A.5, Table A.4; MVSEC [35]. Xie and Piatkowska rows are local adaptations.

From event changes to well-posed estimators

Temporal: control an unobserved baseline.

Spatial: combine directional constraints globally.

Binocular: separate inner uniqueness from correspondence.

The common mathematical question is:

Which directions do the observations control, and which are selected by the model?

Thank you. Discussion and additional proofs follow.

How big is the temporal monocular problem?

Unknowns: HW independent pixel histories, each with N_t samples.

$$\underbrace{(\mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda})}_{\text{SPD, tridiagonal}}\mathbf{f}_{xy} = \mathbf{A}^\top\mathbf{y}_{xy}, \qquad \boxed{\text{solve work } O(HWN_t)}.$$

Work and storage

- Weights, right-hand side, and solve: $O(N_t)$ per pixel.
- Full output: $O(HWN_t)$ storage; solver scratch: $O(N_t)$ per pixel processed at once.
- E events, at most k bins/event: $O(Ek + HWN_t)$ with direct bin indexing.

For a low-power device

- Newest endpoint only: $O(HW)$ work per interval and $O(HW)$ state.
- Bin directly; skip histories with $\mathbf{y}_{xy} = 0$; reuse buffers.
- ROI, coarser sampling, and lower precision trade coverage, detail, or numerical accuracy for cost.

Scale example: $260 \times 346 \times 50 \approx 4.50$ million unknowns; float32 output alone ≈ 18 MB.

Dissertation: Eq. 2.7, Eq. 2.9; edge solver. Uniform bins, fixed kernel order; streaming uses a growing prefix.

References (1 of 3)

[1] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, and Davide Scaramuzza. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1):154–180, 2022.

[2] Harbir Antil and David Sayre. Bilevel inverse problems in neuromorphic imaging. *Inverse Problems*, 39(9):094003, August 2023.

[3] Guillermo Gallego, Henri Rebecq, and Davide Scaramuzza. A unifying contrast maximization framework for event cameras, with applications to motion, depth, and optical flow estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3867–3876, 2018.

[6] Zelin Zhang, Anthony J. Yezzi, and Guillermo Gallego. Formulating event-based image reconstruction as a linear inverse problem with deep regularization using optical flow. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, 45(07):8372–8389, 2023.

[12] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. Events-to-video: Bringing modern computer vision to event cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3857–3866, 2019.

[16] Berthold KP Horn and Brian G Schunck. Determining optical flow. *Artificial Intelligence*, 17(1-3):185–203, 1981.

[17] Harbir Antil, Daniel Blauvelt, and David Sayre. Dynamic reconstruction from neuromorphic data. *Foundations of Data Science*, 2026. Early access. https://doi.org/10.3934/fods.2026017.

[20] Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck. A 128× 128 120 db 15 μs latency asynchronous temporal contrast vision sensor. *IEEE Journal of Solid-State Circuits*, 43(2):566–576, 2008.

[26] Lawrence C. Evans. *Partial differential equations*. American Mathematical Society, 2010.

[27] Andrey Nikolayevich Tikhonov and Vasiliy Yakovlevich Arsenin. *Solutions of Ill-Posed Problems*. V.H. Winston, 1977.

[32] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, MD, 4th edition, 2013.

[35] Alex Zihao Zhu, Dinesh Thakur, Tolga Özaslan, Bernd Pfrommer, Vijay Kumar, and Kostas Daniilidis. The multivehicle stereo event camera dataset: An event camera dataset for 3D perception. *IEEE Robotics and Automation Letters*, 3(3):2032–2039, 2018.

[36] Guillermo Gallego and Davide Scaramuzza. Accurate angular velocity estimation with an event camera. *IEEE Robotics and Automation Letters*, 2(2):632–639, 2017.

[37] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. High speed and high dynamic range video with an event camera. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(6):1964–1980, 2021.

References (2 of 3)

[38] Matthew G. McHarg, Richard L. Balthazor, Brian J. McReynolds, David H. Howe, Colin J. Maloney, Daniel O'Keefe, Rayomand Bam, Gabriel Wilson, Paras Karki, Alexandre Marcireau, and Gregory Cohen. Falcon neuro: an event-based sensor on the international space station. *Optical Engineering*, 61(8):085105, 2022.

[39] Alex Zihao Zhu, Liangzhe Yuan, Kenneth Chaney, and Kostas Daniilidis. Unsupervised event-based learning of optical flow, depth, and egomotion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 989–997, 2019.

[40] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[41] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268, 1992.

[45] Harbir Antil, Daniel Blauvelt, and David Sayre. ESAR: Event-based synthetic aperture reconstruction. arXiv:2607.15073 [eess.IV], 2026.

[46] Yuhuang Hu, Shih-Chii Liu, and Tobi Delbruck. v2e: From video frames to realistic DVS events. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1312–1321, 2021.

[49] Ryad Benosman, Sio-Hoi Ieng, Charles Clercq, Chiara Bartolozzi, and Mandyam Srinivasan. Asynchronous frameless event-based optical flow. *Neural Networks*, 27:32–37, 2012.

[50] Ryad Benosman, Charles Clercq, Xavier Lagorce, Sio-Hoi Ieng, and Chiara Bartolozzi. Event-based visual flow. *IEEE Transactions on Neural Networks and Learning Systems*, 25(2):407–417, 2014.

[55] Shintaro Shiba, Yoshimitsu Aoki, and Guillermo Gallego. Secrets of event-based optical flow. In *European Conference on Computer Vision (ECCV)*, pages 628–645, 2022.

[57] Joachim Weickert and Christoph Schnörr. A Theoretical Framework for Convex Regularizers in PDE-Based Computation of Image Motion. *International Journal of Computer Vision*, 45(3):245–264, 2001.

[66] David Marr and Shimon Ullman. Directional selectivity and its use in early visual processing. *Proceedings of the Royal Society of London. Series B. Biological Sciences*, 211(1183):151–180, 1981.

[67] Hans-Hellmut Nagel and Wilfried Enkelmann. An investigation of smoothness constraints for the estimation of displacement vector fields from image sequences. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(5):565–593, 1986.

[71] Ciyou Zhu, Richard H. Byrd, Peihuang Lu, and Jorge Nocedal. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4):550–560, 1997.

References (3 of 3)

[72] Peter D Lax and Arthur N Milgram. Parabolic equations. In *Contributions to the Theory of Partial Differential Equations*, volume 33 of *Annals of Mathematics Studies*, pages 167–190. Princeton University Press, 1954.

[78] Ewa Piatkowska, Jurgen Kogler, Nabil Belbachir, and Margrit Gelautz. Improved Cooperative Stereo Matching for Dynamic Vision Sensors with Ground Truth Evaluation . In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 370–377. IEEE Computer Society, 2017.

[80] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2nd edition, 2003.

[81] Gilbert Strang. The discrete cosine transform. *SIAM review*, 41(1):135–147, 1999.

[84] Peter J. Green. Iteratively reweighted least squares for maximum likelihood estimation, and some robust and resistant alternatives. *Journal of the Royal Statistical Society: Series B*, 46(2):149–192, 1984.

[87] J. G. Skellam. The frequency distribution of the difference between two Poisson variates belonging to different populations. *Journal of the Royal Statistical Society: Series A*, 109(3):296, 1946.

[92] Luitzen Egbertus Jan Brouwer. Über Abbildung von Mannigfaltigkeiten. *Mathematische Annalen*, 71(1):97–115, 1911.

[95] Heiko Hirschmüller. Stereo processing by semiglobal matching and mutual information. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 30(2):328–341, 2008.

[98] Vaibhav Vaish, Gaurav Garg, Eino-Ville Talvala, Emilio Antunez, Bennett Wilburn, Mark Horowitz, and Marc Levoy. Synthetic aperture focusing using a shear-warp factorization of the viewing transform. In *2006 Conference on Computer Vision and Pattern Recognition Workshop (CVPRW'06)*, pages 129–129. IEEE, 2006.

[103] Paul Rogister, Ryad Benosman, Sio-Hoi Ieng, Patrick Lichtsteiner, and Tobi Delbruck. Asynchronous event-based binocular stereo matching. *IEEE Transactions on Neural Networks and Learning Systems*, 23(2):347–353, 2012.

[104] Jürgen Kogler, Christoph Sulzbachner, Martin Humenberger, and Florian Eibensteiner. Address-event based stereo vision with bio-inspired silicon retina imagers. In Asim Bhatti, editor, *Advances in Theory and Applications of Stereo Vision*, pages 165–188. InTech, 2011.

[105] Zhen Xie, Jianhua Zhang, and Pengfei Wang. Event-based stereo matching using semiglobal matching. *International Journal of Advanced Robotic Systems*, 15(1):1729881417752759, 2018.

[109] Wolfe et al., *Sensation and Perception*, 4th ed. Sinauer Associates, © 2015. Activity 8.2, Movies 5–6 (aperture problem). Publisher animations used unchanged.